VISION Resistivity Dual Frequency Recorded Mode

Schlumberger

Company:	Equinor
Well:	31/5-7
Field:	Eos
Rig Name:	West Hercules
Country:	Norway

Latitude:	60° 34' 35.14" N	UWID:	
Longitude:	3° 26' 36.13" E	Rig Name:	West Hercules
Block:	31/5	Rig Type:	Semi-Submersible

FL:	Norwegian North Sea
FL1:	Section: 12 1/4 x 17 1/5 in.
FL2:	Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level
Ground Level: 307.00 m

Acquisition Dates:	19-Dec-2019 -- 20-Dec-2019
Log Interval:	919.00(m) -- 1908.00(m)
Index Types:	Measured Depth
Index Scales:	1:200
Depth Source:	Driller's Depth
Depth Sensor:	3rd Party Depth
Print Type:	Final
Spud Date:	02-Dec-2019

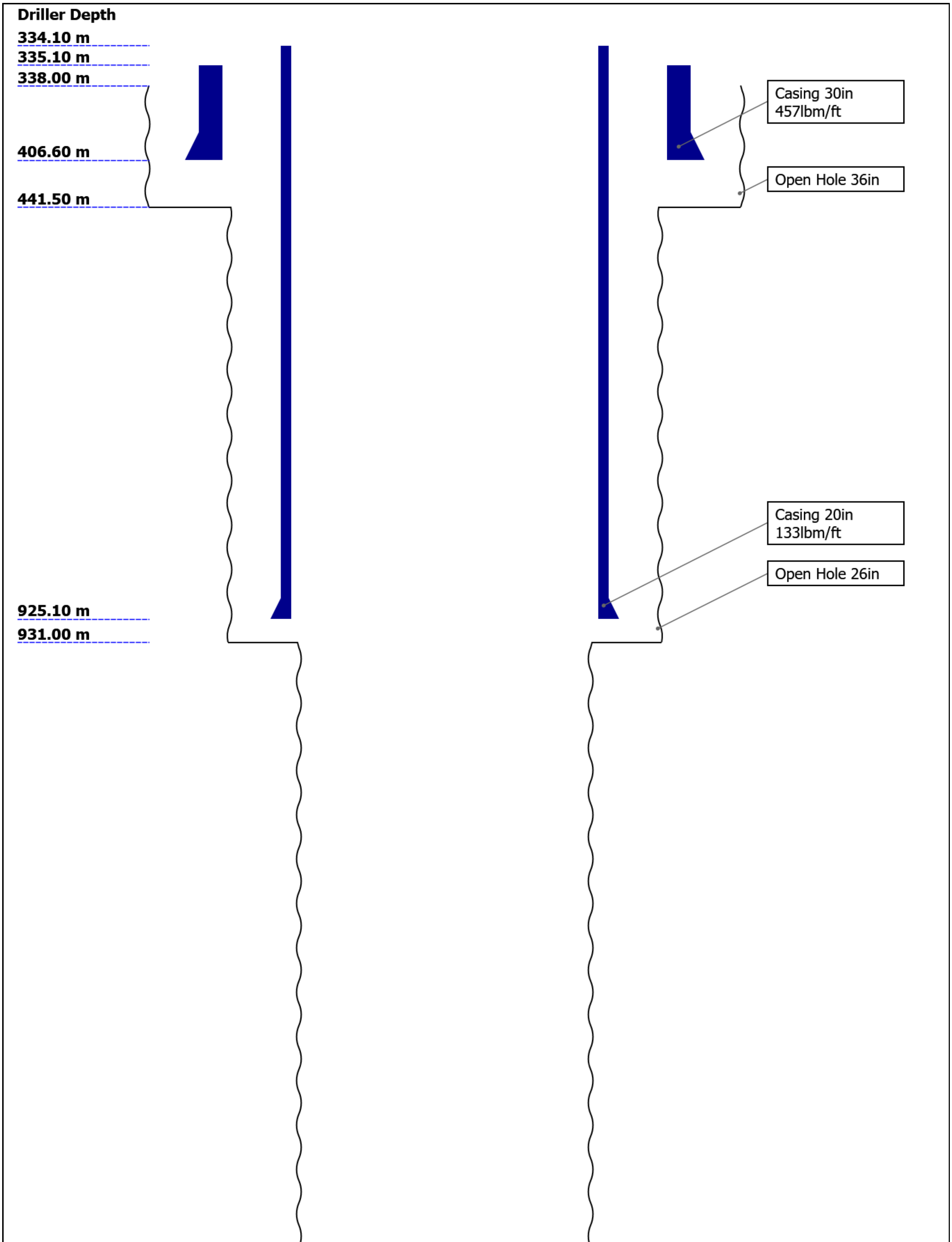
Other Services:
Direction & Inclination (Surveys)
Annular Pressure While Drilling

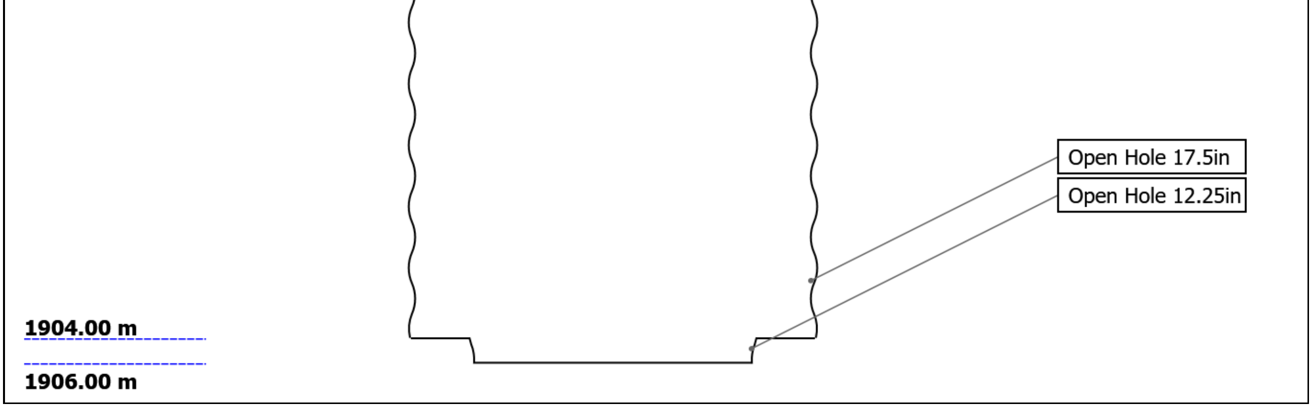
Disclaimer

THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

Contents

Well Sketch


Driller Depth
334.10 m
335.10 m
338.00 m
406.60 m
441.50 m
925.10 m
931.00 m
Casing 30in
457lbm/ft
Open Hole 36in
Casing 20in
133lbm/ft
Open Hole 26in



Borehole Size/Casing Record

Bit							
Bit Size (in)	36	26	17.5	12.25			
Top Driller (m)	338	441.5	931	1904			
Bottom Driller (m)	441.5	931	1904	1906			
Casing							
Size (in)	30	20					
Weight (lbm/ft)	457	133					
Inner Diameter (in)	27.067	18.73					
Grade	X52	N80					
Top Driller (m)	335.1	334.1					
Bottom Driller (m)	406.6	925.1					

Operational Run Summary

Parameter (unit)	**Run 7**						
Date Log Started	19-Dec-2019						
Time Log Started	04:52:35						
Date Log Finished	20-Dec-2019						
Time Log Finished	22:18:15						
Bit Size (in)	17.500						
Bit Start Depth (m)	931.00						
Bit Stop Depth (m)	1906.00						
Top Log Interval (m)	931.00						
Bottom Log Interval (m)	1894.18						
Max Hole Deviation (deg)	1.10						
Azimuth of Max Deviation (deg)	133.27						
Logging Unit Number	A615						
Logging Unit Location	Service Office						
Recorded By	G.Boyd						
Witnessed By	S.Vikra						
Service Order Number	19SCA0085						

Borehole Fluids

Parameter(unit)	Run 7						
Fluid Type	Water						
Fluid Name	Glydril						
Max Recorded Temperatures (degC)	38						
Source of Sample	Active Tank						
Salinity (ppm)	96819.84						
Density (g/cm3)	Zoned						
Funnel Viscosity (s)							
Fluid Loss (cm3)							
PH							
Source RMF							
RMC	Pressed						
RM @ Meas Temp (ohm.m@degC)	0.09 @ 16.3						
RMF @ Meas Temp (ohm.m@degC)	0.06 @ 16						
RMC @ Meas Temp (ohm.m@degC)	0.12 @ 20						
RM @ BHT (ohm.m@degC)	0.06 @ 38						
RMF @ BHT (ohm.m@degC)	0.04 @ 38						
RMC @ BHT (ohm.m@degC)	0.08 @ 38						
Total Solid (%)							
High Gravity Solids (%)							

Zoned Borehole Fluids

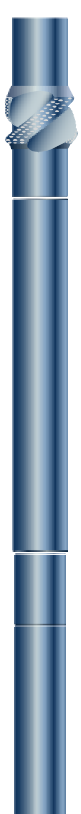
Run 7

Parameter	Value	Start
Density	1.25	12/19/2019 4:52:35 AM
Density	1.26	12/19/2019 5:00:00 PM

Remarks and Equipment Summary

Run 7: Toolstring

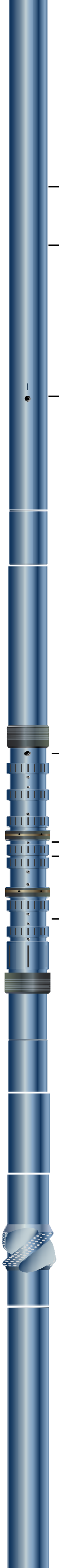
Equip name	Length	MP name	Offset
Stabilizer[2]:26068	28.34	17 1/4 in. Non-Mag Stab	
NMDC: 9 1/2"[2]:26105-1	26.00	Non-Mag Pony Collar	
TELE900:G5093 MSSU900:71539 Upper Extender:26-02-16 MDC900:G5093 MMA:2068 MDI:3956 PMGR:455 PMEA:3708 MTA:2526 MTK900:008/298895-2142 MSSD900:973247	23.35	TeleScope	



Run 7: Remarks

Run 7: Remarks
All depths referenced to driller's depth. See EOWR for depth tracking details.
All data from tool memory. All data acquired while drilling.
Gamma Ray measurement is corrected for collar thickness only and not for Potassium content.
Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
Run 7 and 12 1/4 x 17 1/2 in. section TD at 1906 m.

-1
Lower Extender:1
2/142



D&I 18.78

GR 18.13

ROP 16.43

ARC9:5969 **14.67** arcVISION
ARDC:5969
Upper Extender:0
7-184
CDJA
AREA:8374
APWD:1820
ARSS:105209
Lower Extender:1
3/013

ROP 12.4

GR 11.33
Resistivity 11.28

Pressure 10.57

X/O: 9 1/2":80 990 **8.9** X-Over

Stabilizer[1]:MWS 3529 **8.22** 17 1/4 in. Non-Mag Stab

NMDC: 9 1/2"[1]:25002 **5.98** Non-Mag Pony Collar

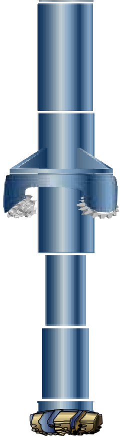
Flt Sub: 9 1/2" :25003 **4.00** Float sub

Hole Opener:JK 4951 **3.34** 17 1/2 in. Hole Opener

X/O: 6 3/4":IOT 25162 **1.69** X-Over

X/O: 8 1/4":OWS-BS-532 **1.28** X-Over

Bit: 12 1/4" **0.3** Smith PDC MDSiZ616



TOOL_ZERO

Lengths are in m
Maximum Outer Diameter = 17.500 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

Run 7

VISION Resistivity - Dual Frequency

Software Version

Acquisition System	Version
Maxwell 2019	9.0.106845.3100
Application Patch	DnM_Hotfix-WITSMLUnit-2019.0_9.0.112817

Computation	Description	Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family	9.0.106845.3100
ARC9GammaRayComputation	ARC9 Gamma Ray Computation Package for both Real-time and Recorded Mode	9.0.106845.3100

Tool Interface	System Version	Loaded Version
HSPM	2019.0b.295	9.0.106845.3100

Tool Elements	Description	Software Version	Firmware Version
DRILLING_SURFACE	DRILLING_SURFACE	9.0.106845.3100	
ARDC	ARC 9.00 Inch Tool Drilling Collar	9.0.106845.3100	V9.7B

Pass Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 7	Drilling	Down	34.44 m	1905.58 m	19-Dec-2019 4:52:35 AM	20-Dec-2019 10:18:15 PM	Yes

All depths are referenced to toolstring zero

Log

Company:Equinor Well:31/5-7

Run 7: Drilling:S031

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 23-Dec-2019 16:36:11

TICK_ARC_RES - Resistivity Tick Marks ARC9 RM

TICK_ARC_GR - Gamma Ray Tick Marks ARC9 RM

Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM

Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM

Phase Shift Resistivity 34 inch

Resistivity Time After Bit (ARTM) ARC9
0 h 10

Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT
200 m/h 0

Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM
0 gAPI 150

Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM
150 gAPI 300

0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM
0.2 ohm.m 200

0.2 ohm.m 200

Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM
0.2 ohm.m 200

Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM
0.2 ohm.m 200

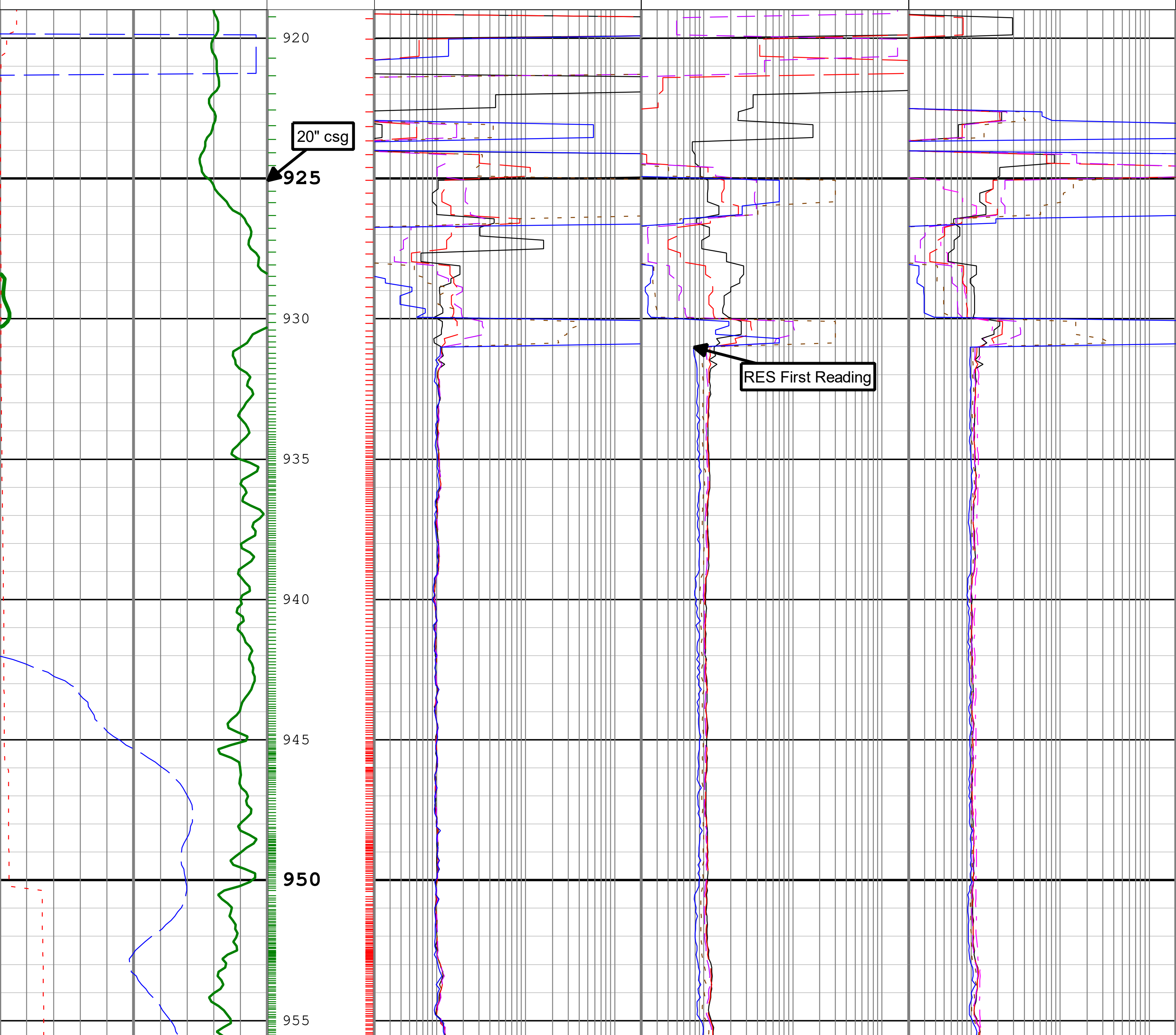
Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM
0.2 ohm.m 200

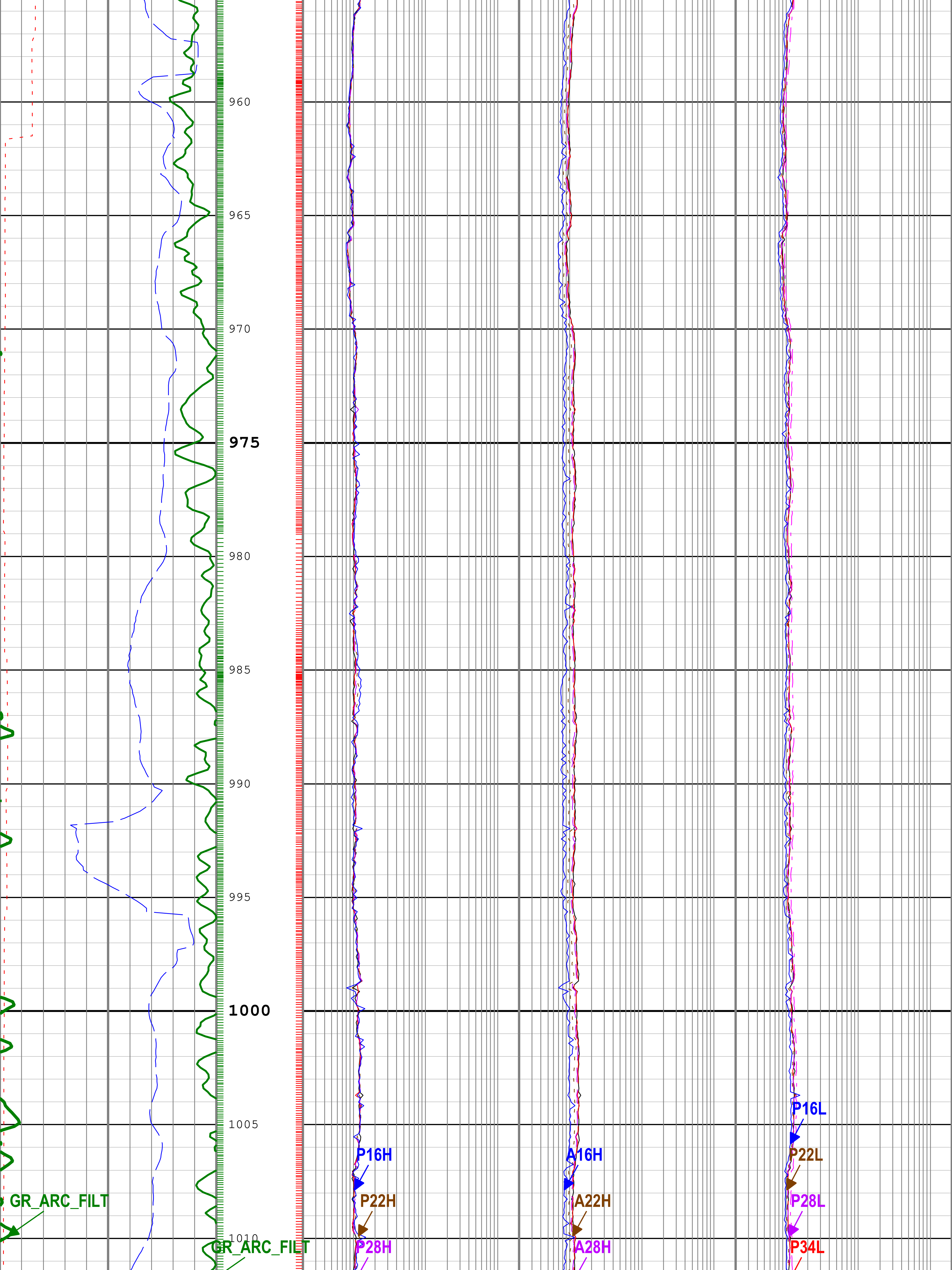
Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM
0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM
0.2 ohm.m 200




960
965
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985
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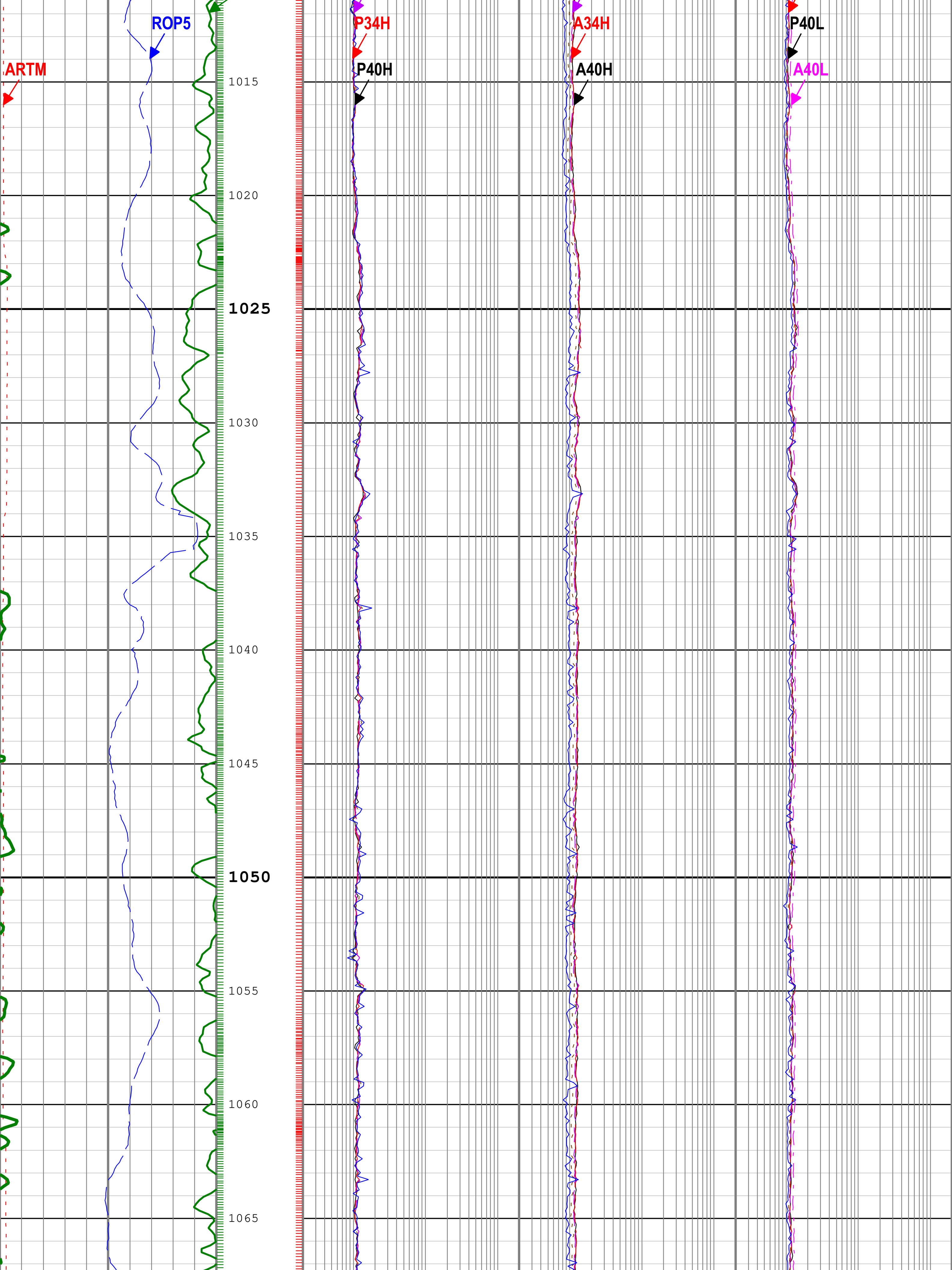
995
1000
1005
1010
GR_ARC_FILT
GR_ARC_FILT
P16H
P22H
P28H
A16H
A22H
A28H
P16L
P22L
P28L
P34L


ROP5
ARTM
P34H
P40H
A34H
A40H
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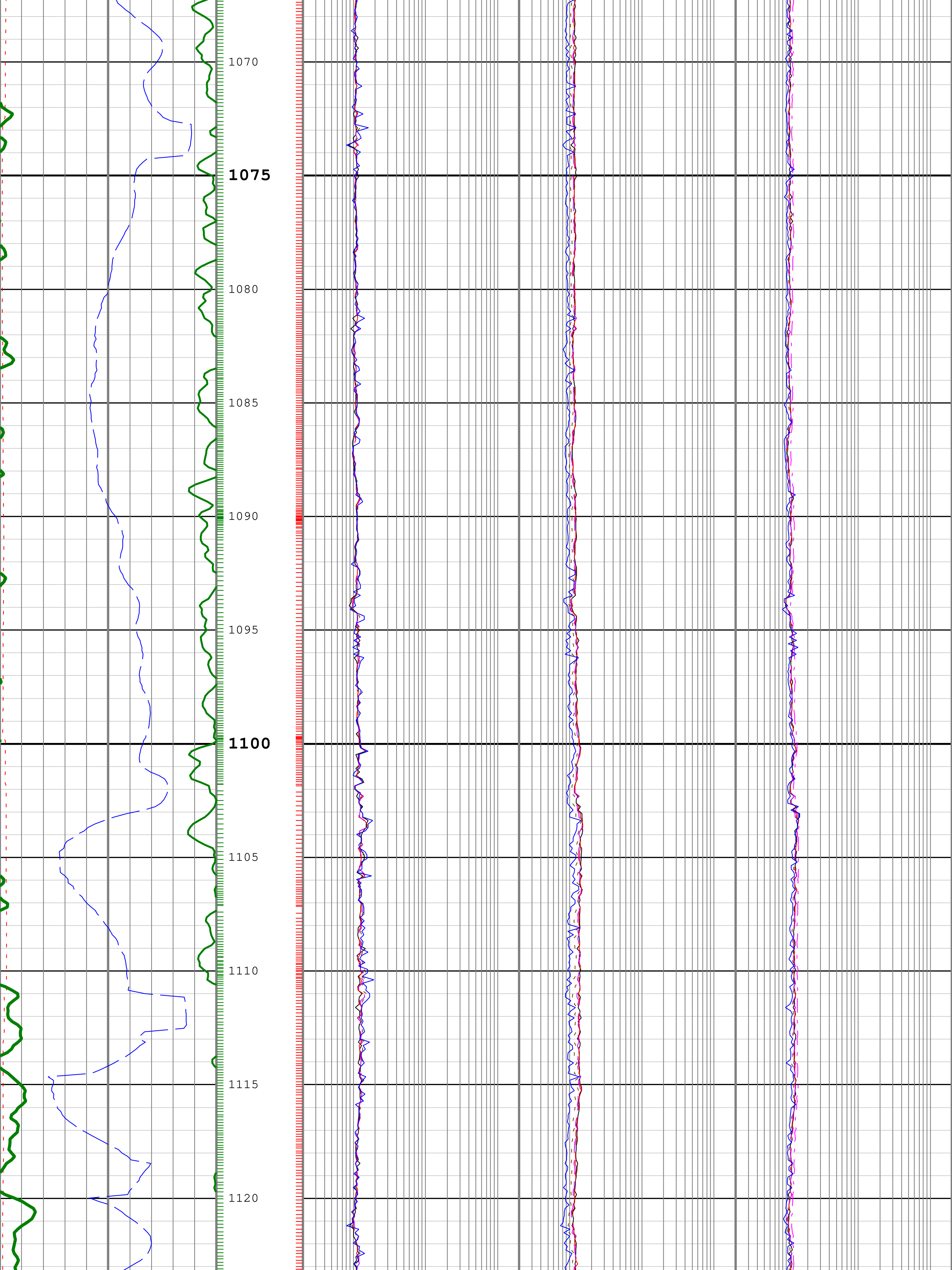
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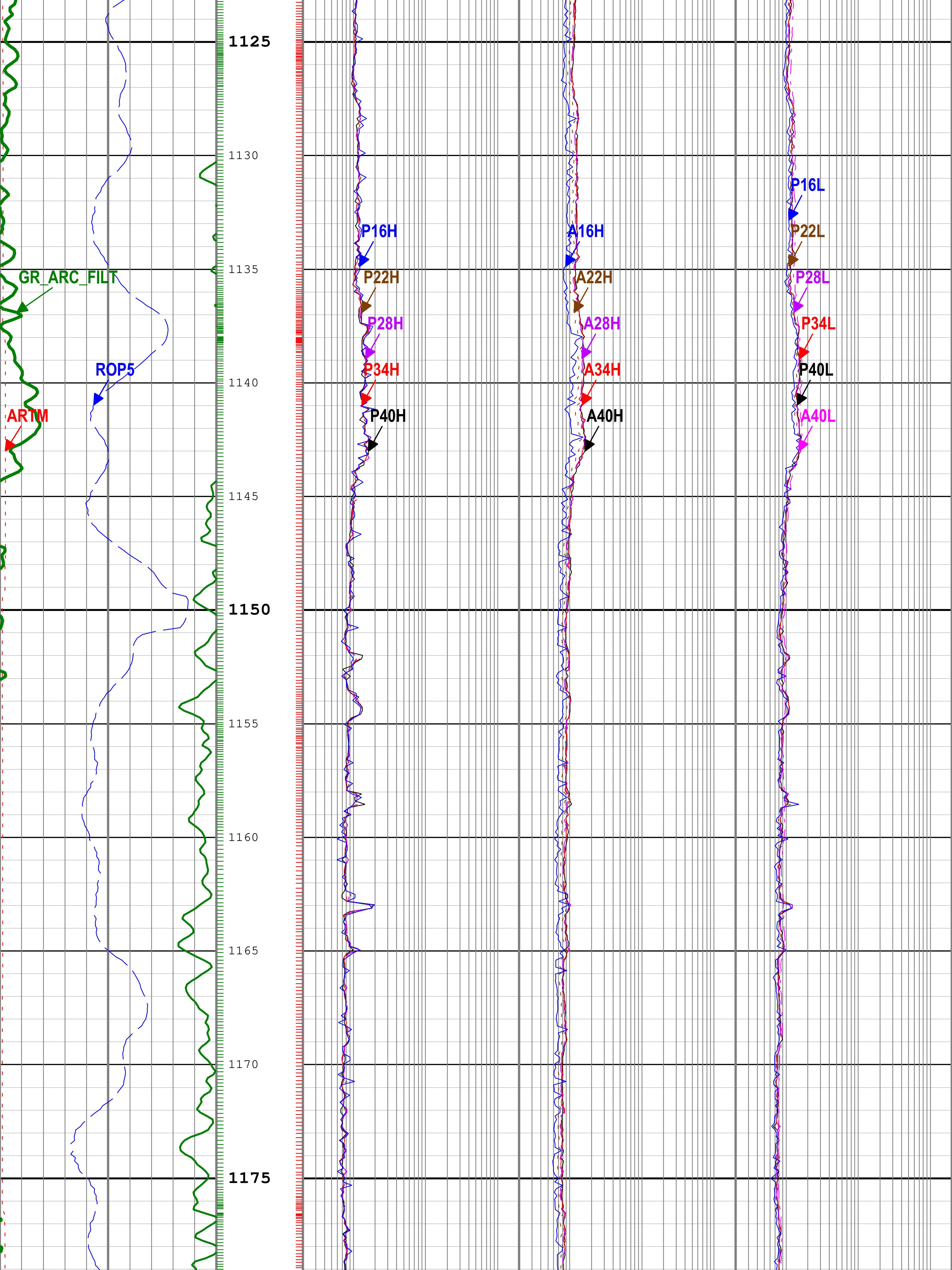
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GR_ARC_FILT
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P16H
P22H
P28H
P34H
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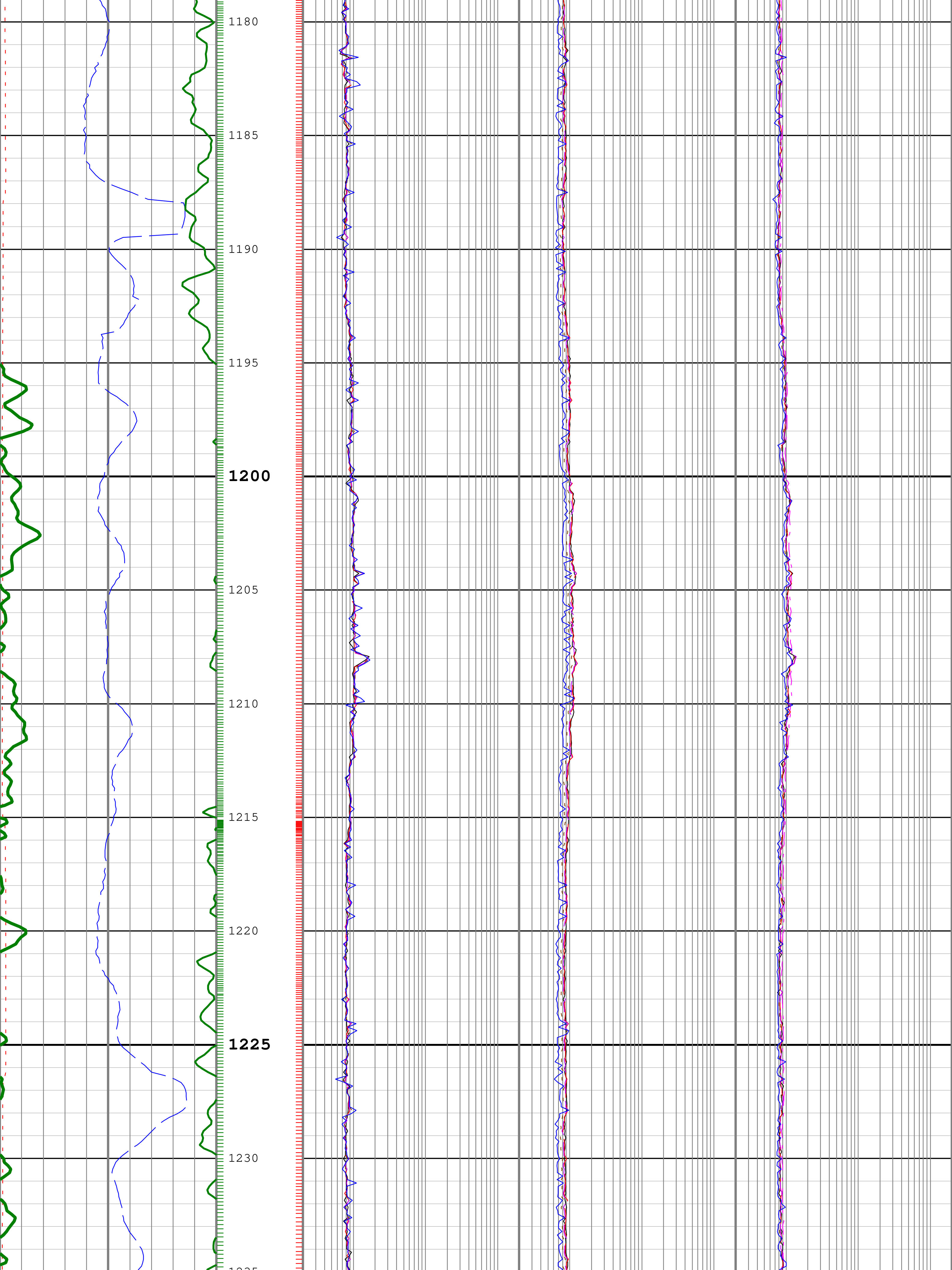
P34L
P40L
A40L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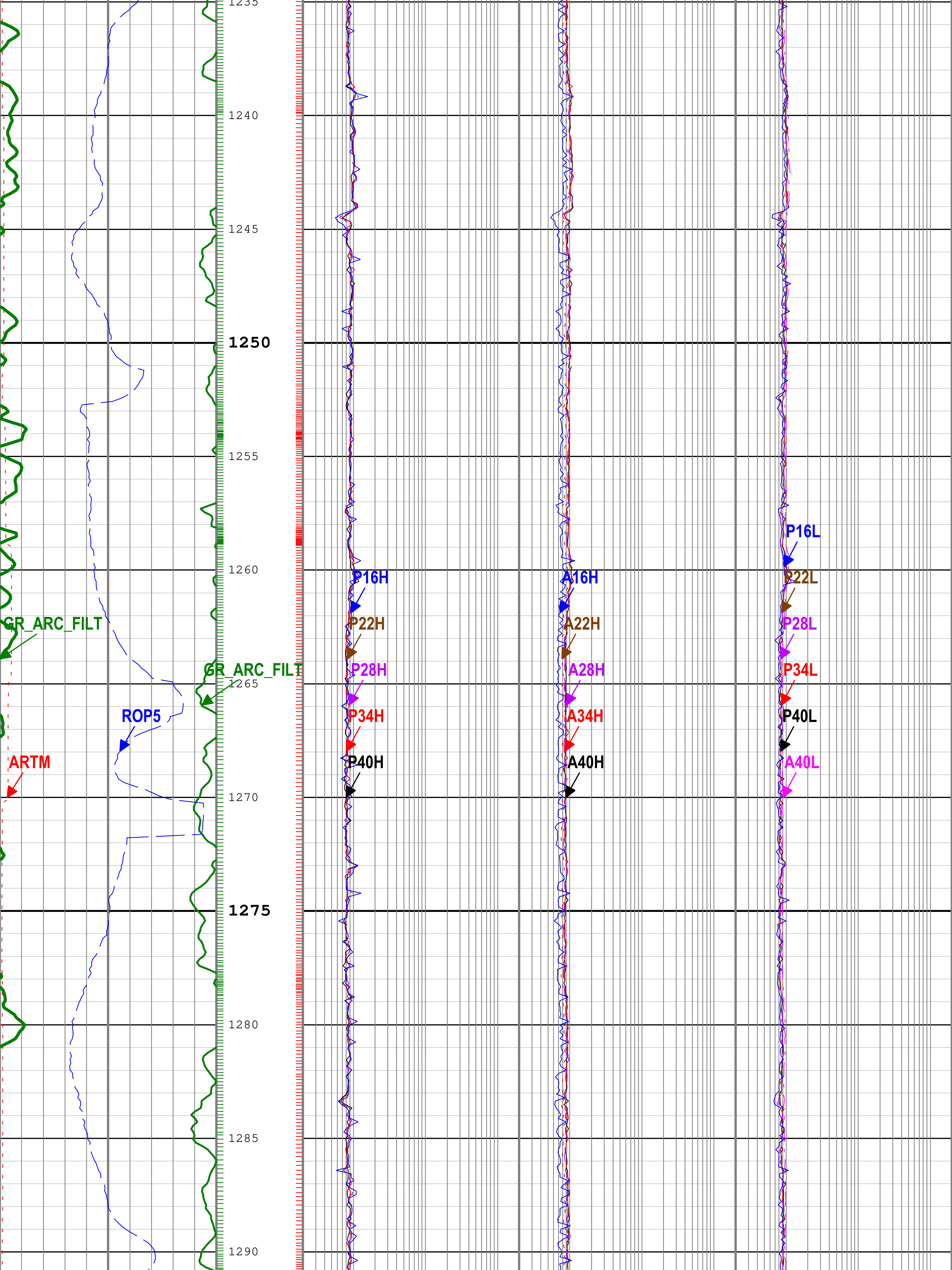
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ROP5
ARTM
GR_ARC_FILT
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P16H
P22H
P28H
P34H
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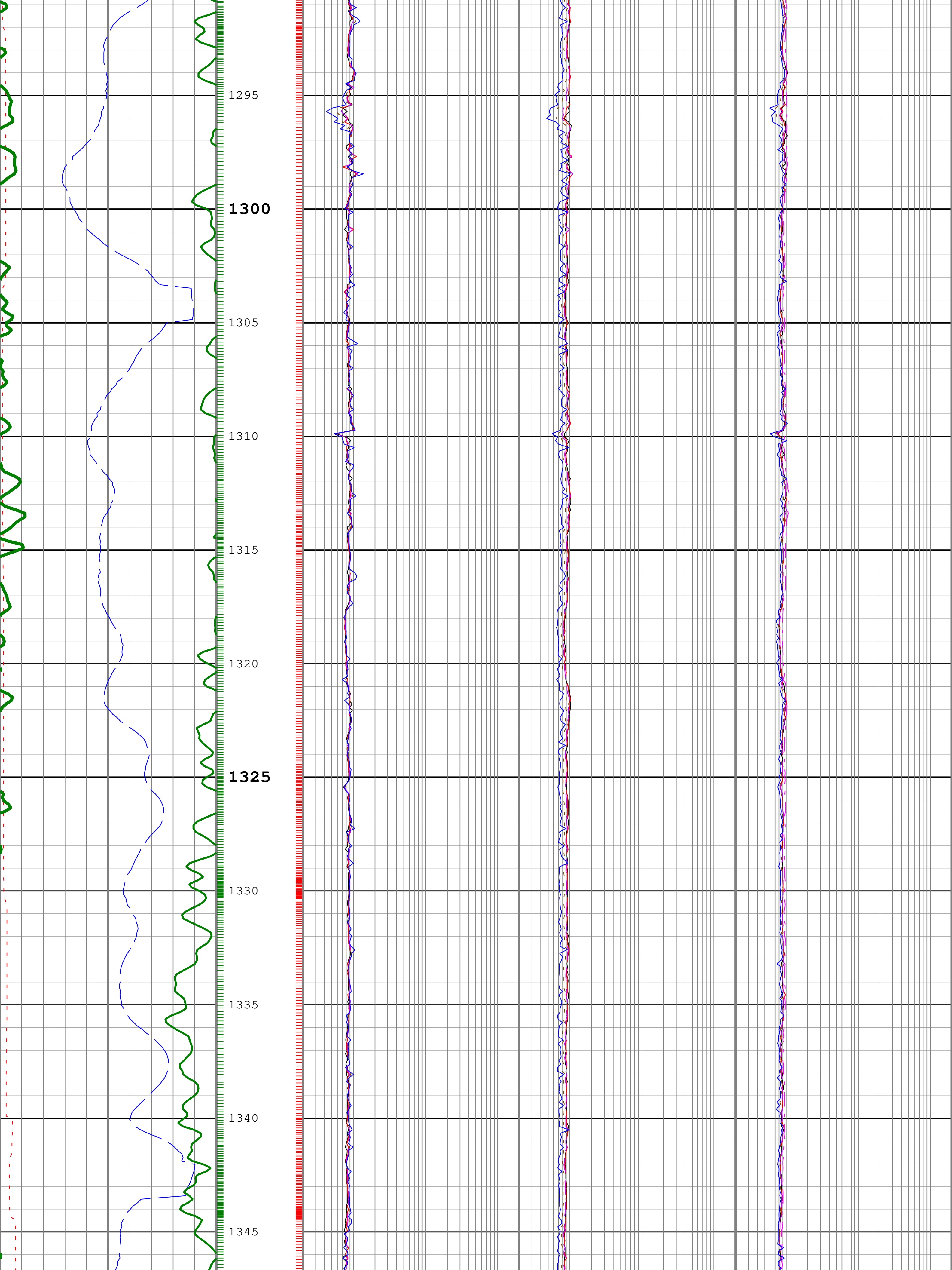
A40L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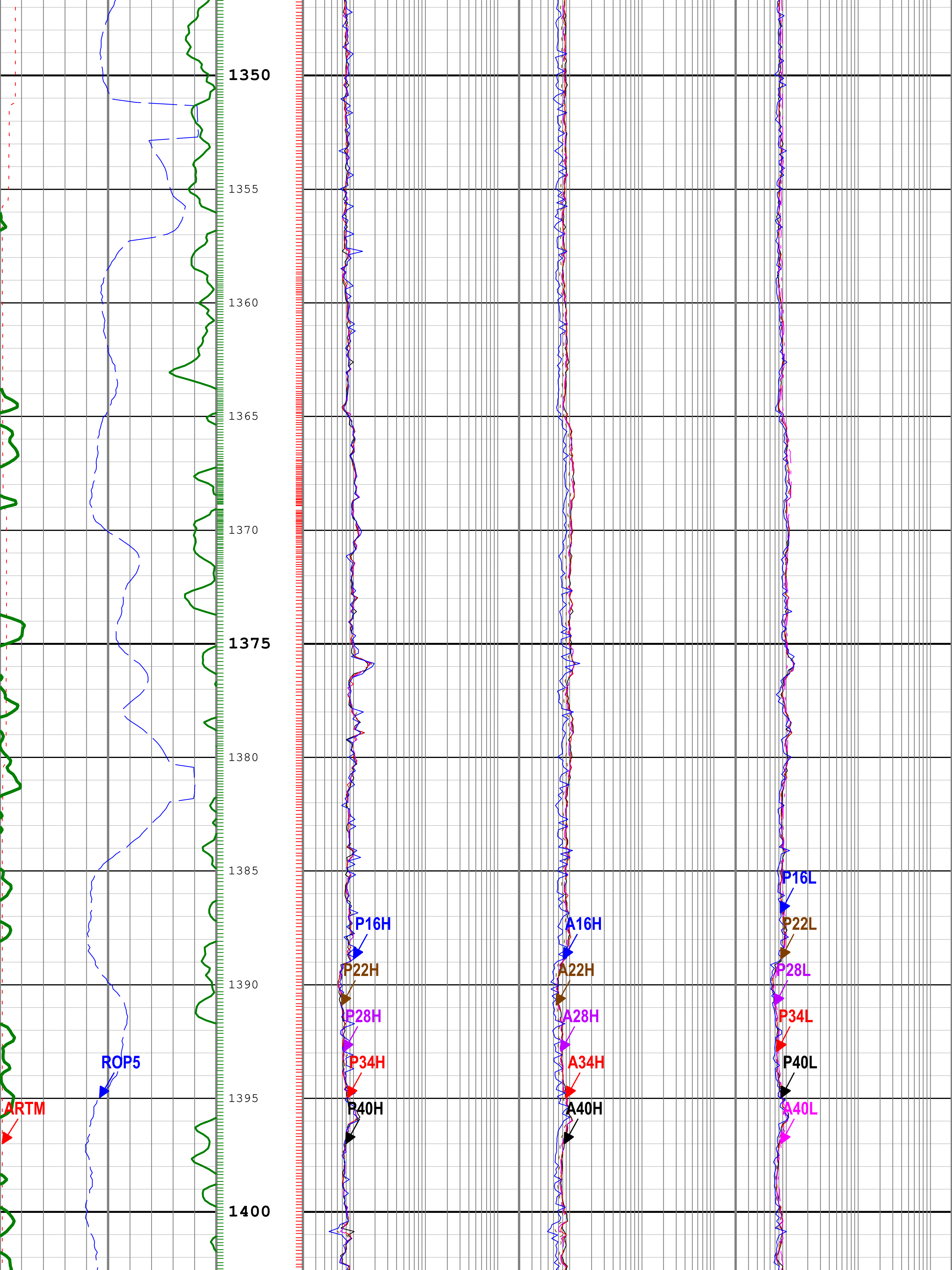
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P28H
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P34L
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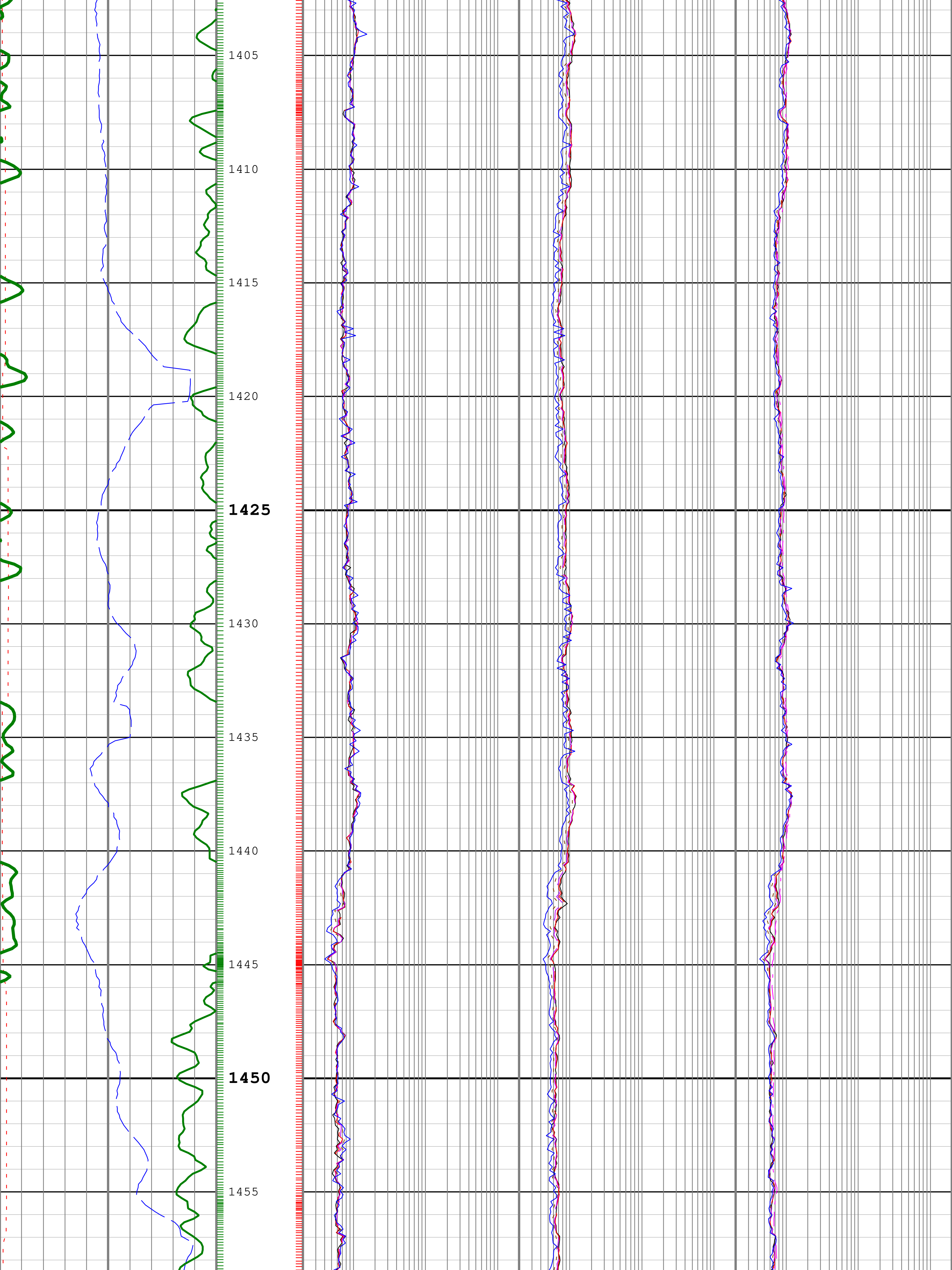
A40L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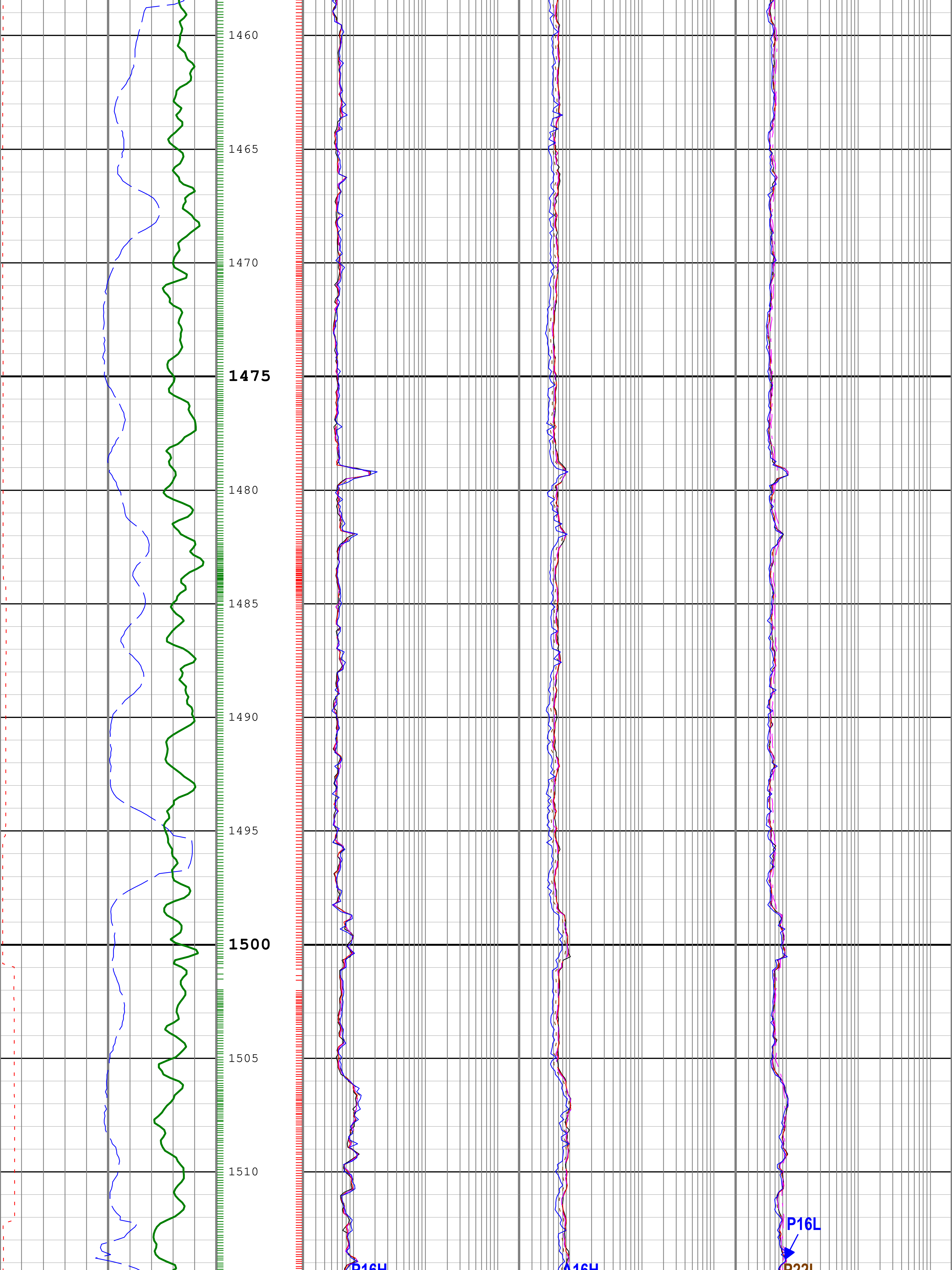
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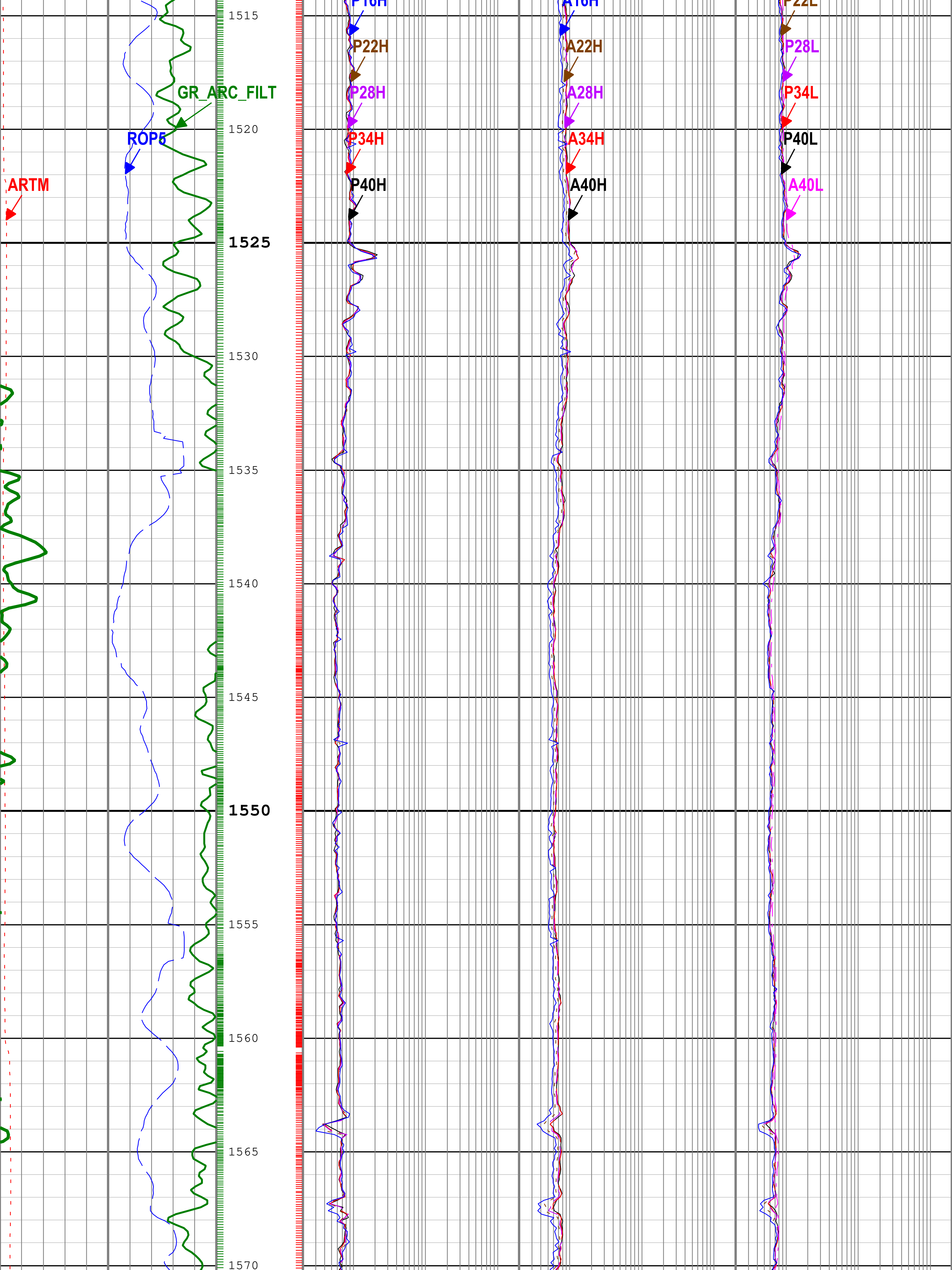
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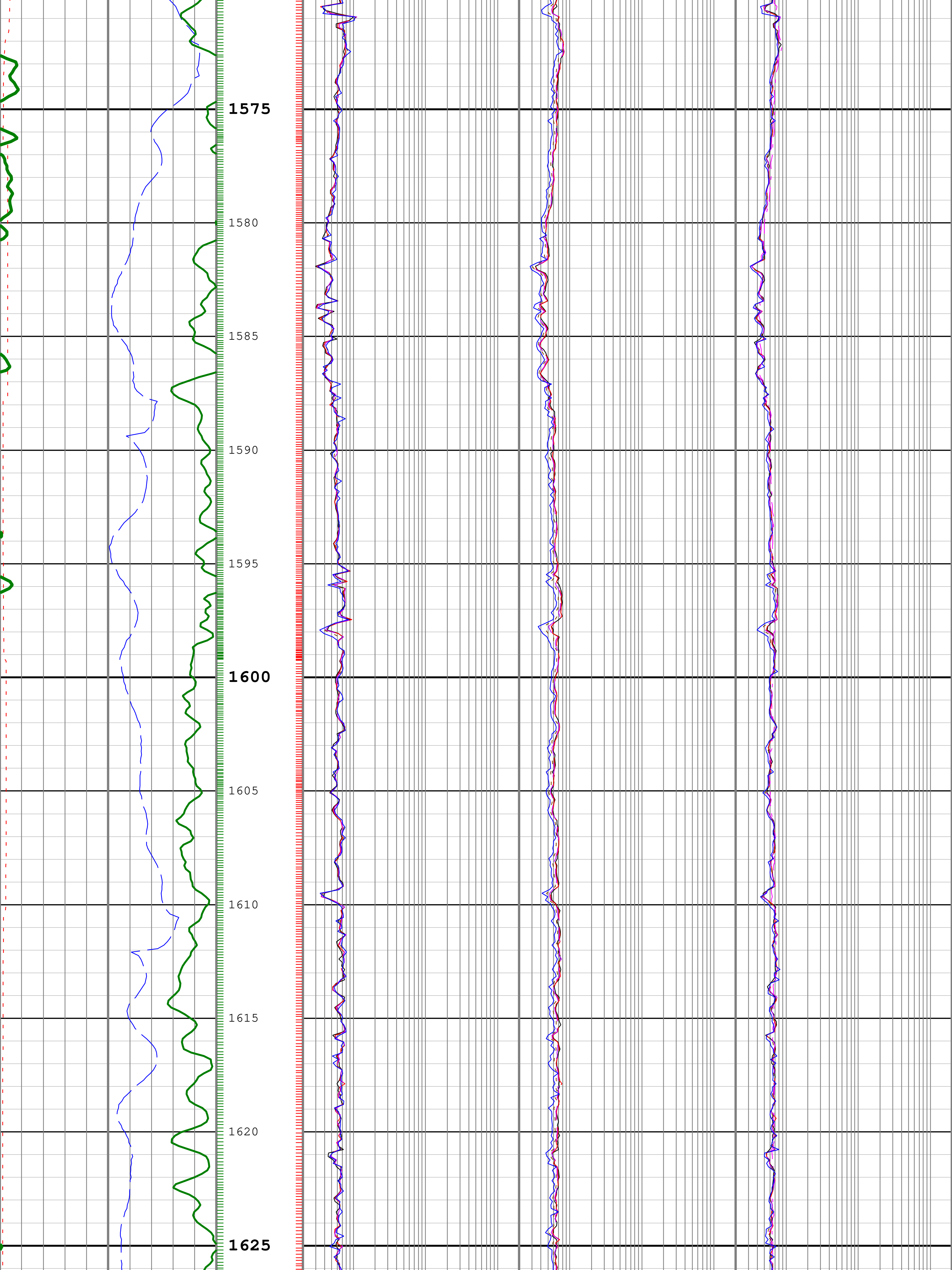
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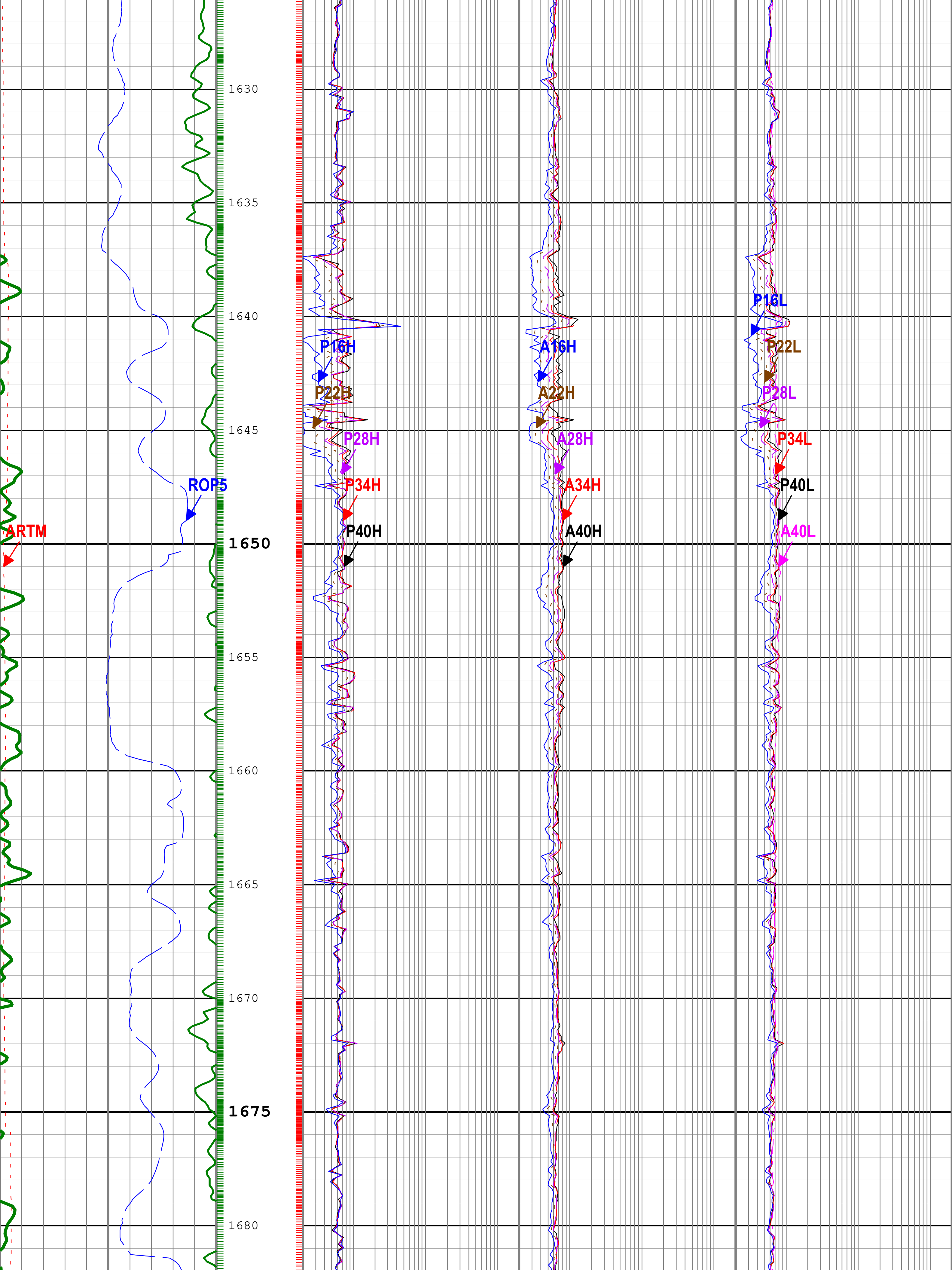
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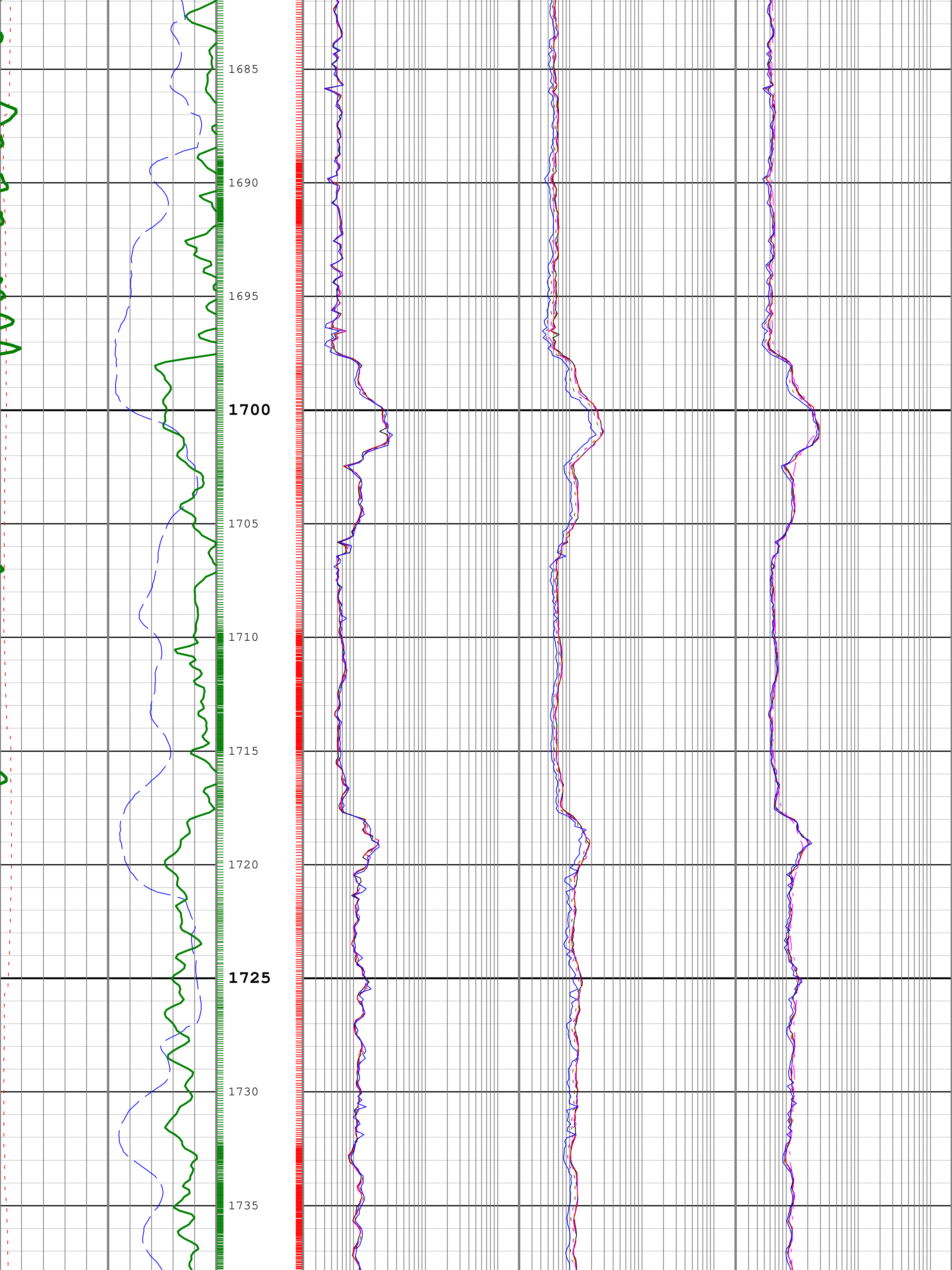
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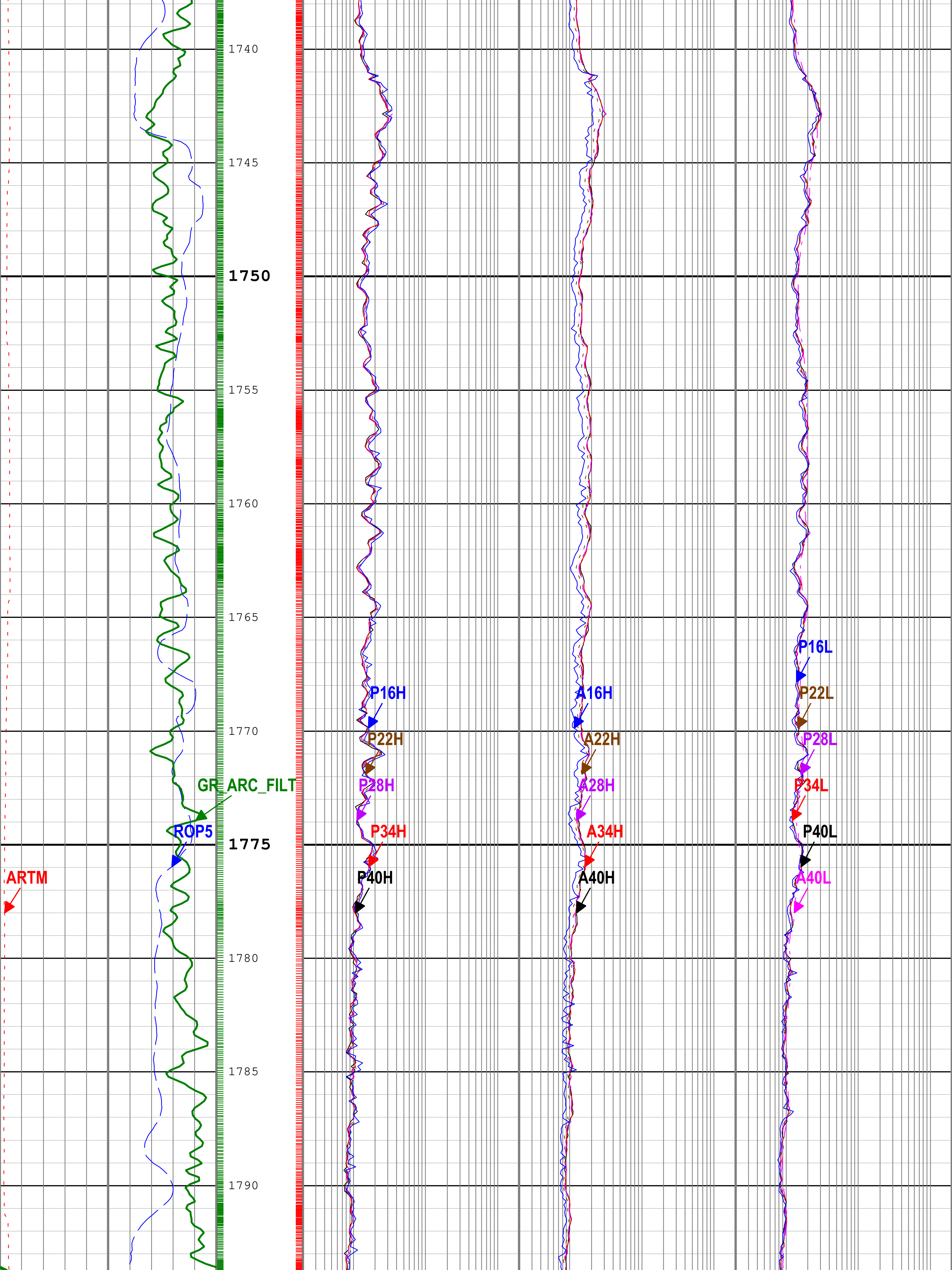
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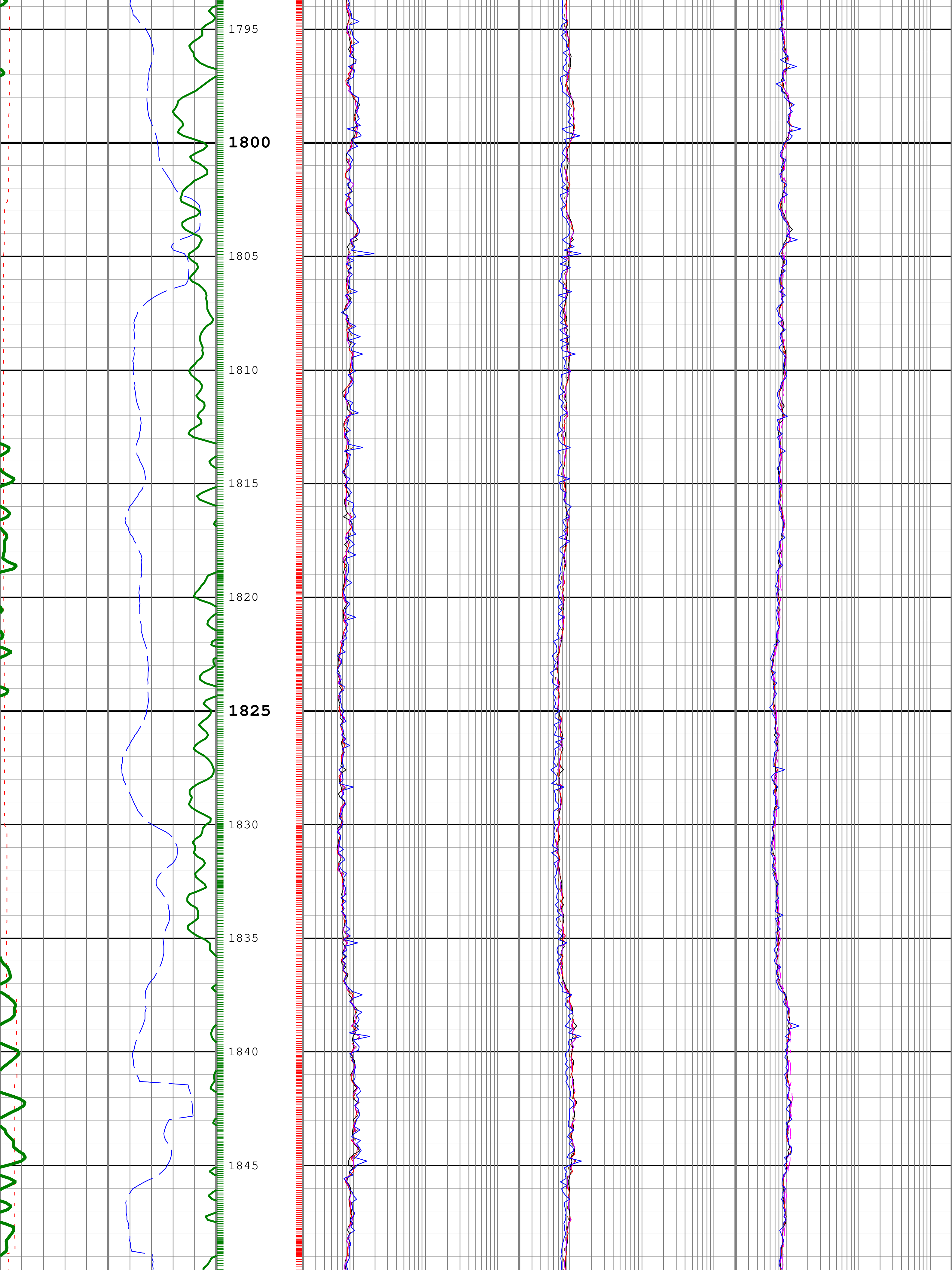
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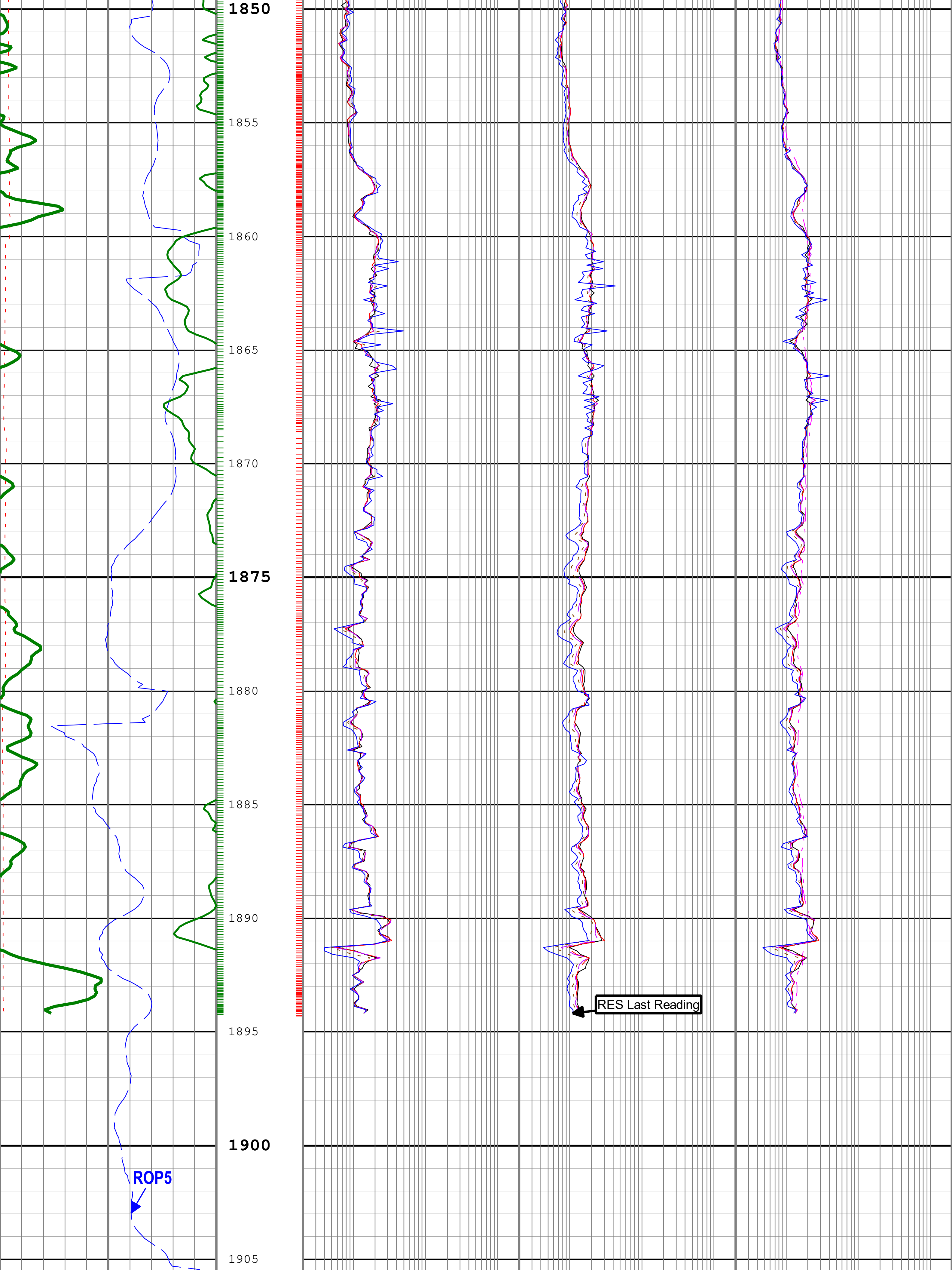
1850
1855
1860
1865
1870
1875
1880
1885
1890
RES Last Reading
1895
1900
ROP5
1905

TD @ 1906 m

Resistivity Time After Bit (ARTM) ARC9

0 h 10

Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT

200 m/h 0

Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM

0 gAPI 150

Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM

150 gAPI 300

Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM

0.2 ohm.m 200

Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM

0.2 ohm.m 200

Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM

0.2 ohm.m 200

Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM

0.2 ohm.m 200

Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM

0.2 ohm.m 200

Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM

0.2 ohm.m 200

Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM

0.2 ohm.m 200

Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM

0.2 ohm.m 200

TICK_ARC_GR - Gamma Ray Tick Marks ARC9 RM

TICK_ARC_RES - Resistivity Tick Marks ARC9 RM

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 23-Dec-2019 16:36:11

Channel Processing Parameters

Run 7: Parameters

Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC9	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	0	%
BS	Bit Size	DNMSESSION	Depth Zoned	in
DFD	Drilling Fluid Density	Borehole	Time Zoned	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC9	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC9	1	ohm.m
RMS	Resistivity of Mud Sample	Borehole	0.09	ohm.m

Depth Zone Parameters

Parameter	Value	Start (m)	Stop (m)
BS	26	919	931
BS	17.5	931	1904

BS	12.25	1904	1906

All depth are actual.

Time Zone Parameters

Parameter	Value	Start Time	Stop Time	Start Depth (m)	Stop Depth (m)
DFD	1.25	19-Dec-2019 04:52:35	19-Dec-2019 17:00:00	34.44	1264.82
DFD	1.26	19-Dec-2019 17:00:00	20-Dec-2019 22:18:15	1264.82	1905.58

All depth are at tool zero.

Tool Control Parameters

Calibration Report

ARC9 (Array Resistivity Compensated 900) Calibration - Run 7

Primary Equipment :

High Pressure without AIM Receiver AREA 8374

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 09:51:54 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	8.353	10.500
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.149	8.500
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	5.129	6.500
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.142	6.600
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.746	5.600
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	0.713	4.100
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-0.663	4.100
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	0.666	4.100
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-0.695	4.100
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	0.651	4.100
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	8.523	10.500
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	5.988	8.500
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	5.294	6.500
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	3.975	6.600
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.915	5.600
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	0.360	4.100
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	-0.415	4.100
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	0.390	4.100
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	-0.422	4.100
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	0.384	4.100

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:35:50 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit
Gamma Ray Calibration Gain		Master	1.000	0.640	0.979	1.200

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Country: Norway

Schlumberger

VISION Resistivity

Dual Frequency

Recorded Mode